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SMART VEHICLE SECURITY SYSTEM USING RFID & KEYPAD

ABSTRACT

This project presents the design and implementation of a Vehicle Security System using RFID and Keypad Authentication to enhance protection against unauthorized vehicle access. The system integrates an RFID (Radio Frequency Identification) reader module, Arduino microcontroller, keypad interface, LCD display, relay module, buzzer, and DC motor to create a multi-layer authentication mechanism.

In the proposed system, the RFID module is primarily used for driver authentication. When a valid RFID tag is detected, the microcontroller verifies the stored identification data and grants access by activating the relay that controls the vehicle ignition (represented by the DC motor). If the RFID tag is invalid, the system triggers a buzzer alarm and denies access.

To ensure system reliability and flexibility, a manual authentication method using a 4-digit keypad password is implemented as a secondary verification mechanism in case the RFID card is unavailable. The user must enter the correct password through the keypad interface. If the entered password matches the predefined credentials stored in the microcontroller memory, the system authorizes ignition and the motor starts. Otherwise, the system activates the buzzer to indicate unauthorized access.

The LCD display module provides real-time system status messages such as authentication prompts, access granted, or access denied. The integration of dual authentication techniques improves security, reliability, and user accessibility compared to traditional vehicle locking systems.

Overall, this embedded system demonstrates an effective low-cost and efficient anti-theft solution that combines RFID-based identification and password-based verification for enhanced vehicle protection.

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